

The Greenholt Phish

SOC Analyst Tier 1 · Phishing Email Analysis · Attachment Forensics

Phishing

BEC / Wire Transfer

Header Analysis

SPF / DMARC

VirusTotal

1 Scenario

A sales executive flagged a suspicious email requesting a money transfer. The email had a generic greeting and an unusual tone. The task was to deconstruct the email headers, verify domain security posture, and assess the attachment for malicious intent.

2 Initial Triage — Email Metadata

Subject / Reference	Sender Display Name	From Address	Reply-To Address
Transfer ref: 09674321	Mr. James Jackson	info@mutawamari ne.com	info.mutawamarin e@mail.com

▲ The Reply-To address uses a different domain (mail.com) from the sender domain (mutawamarine.com). This is a classic reply-hijacking tactic — any response from the victim goes directly to the attacker.

3 Infrastructure Investigation — Originating IP

X-Originating-IP	ISP / Owner	Expected Mail Host
192.119.71.157	Hostwinds LLC	Outlook (Microsoft)

✗ The email originated from a Hostwinds LLC IP — a commercial hosting provider — while the domain's SPF record authorizes only Microsoft/Outlook mail servers. This mismatch is a strong indicator of domain spoofing.

4 Domain Security Check — SPF & DMARC

SPF Record

```
$ nslookup -type=txt mutawamarine.com
v=spf1 include:spf.protection.outlook.com -all
# Only Outlook IPs are authorized – Hostwinds is NOT listed
```

DMARC Record

```
$ nslookup -type=txt _dmarc.mutawamarine.com
v=DMARC1; p=quarantine; fo=1
# p=quarantine: failures go to spam – NOT rejected at gateway
```

▲ A p=quarantine policy is better than none, but p=reject would have blocked this email outright at the gateway level. Recommend tightening enforcement.

5 Attachment Forensics — Malware Analysis

Filename	Declared Extension	True File Type	File Size
SWT_#09674321_ ____PDF____.CAB	.CAB (Cabinet archive)	RAR archive	400.26 KB

SHA-256 Hash

```
$ sha256sum attachment.CAB
2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f
```

✗ VirusTotal confirmed the hash as malicious. The file masquerades as a .CAB Windows cabinet but is actually a RAR archive — a deliberate mislabeling technique to bypass content-based filters that check extensions without inspecting magic bytes.

! Key Findings Summary



Reply-To hijacking

Replies routed to attacker-controlled domain (mail.com). Victim would unknowingly communicate with the threat actor.



SPF authentication failure

Originating IP (Hostwinds LLC: 192.119.71.157) is not authorized by mutawamarine.com's SPF record — clear spoofing indicator.



Weak DMARC enforcement

p=quarantine allowed the email to reach the inbox as spam rather than being rejected at the MTA level.



Malicious attachment with spoofed extension

RAR archive disguised as a .CAB file to evade extension-based filtering. Confirmed malicious on VirusTotal (SHA-256 hash match).

VERDICT — CONFIRMED MALICIOUS

This is a textbook Business Email Compromise (BEC) phishing attempt. The convergence of a spoofed sender, mismatched Reply-To, SPF failure, and a mislabeled malicious attachment leaves no ambiguity. Recommend blocking the originating IP (192.119.71.157), escalating to the IR team, and upgrading the domain DMARC policy from p=quarantine to p=reject.

Tools Used

nslookup	sha256sum	VirusTotal	TryHackMe
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